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Research on Retrieval-Augmented Generation Method Based on Dynamic Knowledge Indexing and Semantic Alignment Optimization

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***Abstract*—Large language models (LLMs) face significant challenges in knowledge-intensive tasks, including knowledge solidification, insufficient timeliness, and hallucination generation. To address these challenges, this study proposes an innovative retrieval-augmented generation (RAG) method based on dynamic knowledge indexing and semantic alignment optimization. However, traditional RAG methods suffer from semantic fragmentation and query-knowledge misalignment. The approach constructs a dynamic multi-granularity knowledge indexing system and integrates an intelligent dual-path routing decision mechanism to achieve efficient collaboration between knowledge retrieval and generation processes. Key innovations include: (1) a dynamic multi-granularity (comprising lexical-level inverted indexing and semantic-level vector indexing) knowledge indexing architecture that employs an adaptive chunking algorithm based on semantic coherence analysis to resolve semantic fragmentation caused by fixed-length segmentation; (2) a prompt engineering-driven semantic alignment optimization mechanism that unifies the semantic space of queries and knowledge bases by generating hypothetical question-answer pairs, significantly improving retrieval accuracy. Experimental results demonstrate significant improvements: 37.01 EM and 47.53 F1 on HotpotQA, 35.12 EM and 45.69 F1 on SQuAD. This work provides a reliable technical solution for intelligent question answering and advances RAG technology in semantic understanding and retrieval accuracy.**

Keywords-Retrieval-Augmented Generation; Large Language Model; Dynamic Knowledge Indexing; Semantic Alignment; Intelligent Routing

I. INTRODUCTION

With the proliferation of large language models (e.g., GPT-4, LLaMA), intelligent question-answering systems have demonstrated powerful capabilities in natural language processing tasks. However, LLMs still face challenges such as insufficient knowledge timeliness, frequent hallucination phenomena, and poor adaptability to vertical domains [1-2].

Retrieval-Augmented Generation (RAG) [3-6] is a technique that integrates LLMs with external knowledge bases, aiming to enhance the accuracy and reliability of generated content through dynamic retrieval of relevant information. Its core consists of three stages: indexing, retrieval, and generation. First, a knowledge base index is constructed; then, relevant documents are retrieved based on queries; finally, answers are generated using LLMs. By reducing the reliance of LLMs on parametric knowledge, RAG effectively mitigates the model hallucination problem (i.e., generating false or erroneous information), particularly excelling in knowledge-intensive tasks such as open-domain question answering (ODQA) [7] and vertical domain applications.

Despite its potential, practical applications of RAG face multiple challenges: Traditional retrieval methods (e.g., sparse retrieval) rely on lexical matching, struggling to capture semantic relevance, which may lead to inaccurate or misleading retrieved documents. For instance, as our experimental results in TABLE II show, a traditional RAG baseline achieves only 13.40 EM and 23.10 F1 on the SQuAD dataset, highlighting significant room for improvement in retrieval accuracy. In vertical domains such as education and healthcare, LLMs find it difficult to deeply mine user-specific needs [8], and knowledge updates are slow, unable to respond to dynamic information in real-time. Fine-tuning LLMs or constructing knowledge bases requires substantial annotated data and computational resources, limiting the deployment of RAG in resource-constrained scenarios [9-10]. The grammatical and semantic gaps between user queries and knowledge base passages reduce retrieval efficiency and quality.

This study proposes an innovative dynamic knowledge retrieval enhancement framework, with main contributions including:

(1) Designing a dynamic multi-granularity knowledge indexing architecture that supports real-time parsing of heterogeneous data and hybrid index construction, achieving adaptive chunking through semantic coherence analysis;

(2) Introducing a prompt engineering-driven semantic alignment optimization mechanism that enhances the semantic matching degree between queries and the knowledge base by generating hypothetical question-answer pairs;

(3) Proposing a dual-channel intelligent routing decision model that dynamically selects retrieval and generation paths.

II. RELATED WORK

A. Traditional Retrieval-Augmented Generation Methods

Early RAG research primarily focused on the simple combination of dense passage retrieval (DPR) and generative models. These methods map queries and documents into the same vector space using a dual-encoder structure but suffer from insufficient semantic understanding. Guu et al. [11] proposed a retrieval-augmented language model pre-training method that integrates the retriever into the pre-training process, enhancing the model's knowledge acquisition capability, but at high computational cost. Subsequent studies, such as Qi et al.,

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pointed out that directly storing corpora in segments leads to low semantic matching between queries and segmented content, affecting generation quality.

B. Hybrid Retrieval Strategies

To overcome the limitations of single retrieval methods, researchers have proposed hybrid retrieval approaches that combine the advantages of sparse retrieval (e.g., BM25) and dense retrieval. The DPR method proposed by Karpukhin et al. [12] achieved significant results in open-domain question answering tasks but still lacks precision when handling complex queries. Studies by Jia et al. further indicate that integrating keyword matching and semantic vector retrieval can balance lexical overlap and semantic relevance but requires optimization of the ranking mechanism.

C. Dynamic Knowledge Update Mechanisms

Recent research has begun to focus on the dynamic updating of knowledge bases. The RAG framework proposed by Lewis et al. [3] supports real-time updates of the knowledge base but still has shortcomings in maintaining knowledge consistency. Compared to existing work, the method proposed in this study introduces innovative improvements in index structure, semantic alignment, and routing decisions, particularly addressing semantic fragmentation and query alignment issues through dynamic semantic chunking and prompt engineering optimization.

III. Methodology

A. Overall Architecture

Figure 1 illustrates the comprehensive framework of the proposed dynamic knowledge indexing and semantic alignment optimization for retrieval-augmented generation method. Our proposed method achieves efficient synergy through the deep integration of four architectures: The dynamic multi-granularity indexing architecture adopts a two-level system of lexical-level inverted indexing and semantic-level vector indexing, combined with a cosine similarity-based adaptive chunking algorithm to resolve semantic fragmentation issues; the prompt engineering-driven mechanism generates hypothetical question-answer pairs using the 5W1H framework to unify the semantic space; the intelligent routing model uses a lightweight neural network to fuse multiple features and calculates retrieval probability via a sigmoid function for dynamic path selection; the hybrid retrieval mechanism combines the advantages of BM25 and DPR, ensuring result accuracy through weighted scores and LambdaMART re-ranking. The design principles follow semantic coherence theory for chunking and information retrieval theory for index construction. Parameter selection was based on grid search optimization and empirical validation on validation datasets.

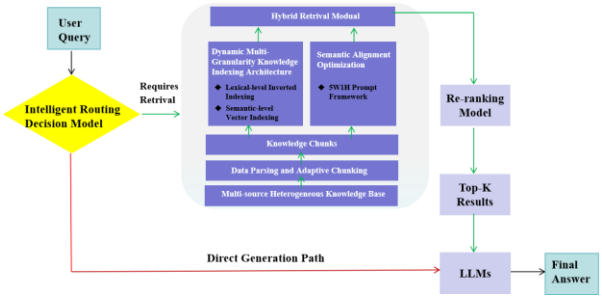


Figure 1. Overall Architecture

B. Dynamic Multi-Granularity Knowledge Indexing Architecture

1) Data Preprocessing and Knowledge Extraction

The proposed method supports multi-source heterogeneous data input, including Word, PDF, HTML, and other formats. A data parser converts various documents into a unified intermediate format, followed by entity extraction, relationship identification, and semantic annotation. The knowledge extraction process uses a BERT-based sequence labeling model. Specific steps include: converting multi-source data into unified formats like Markdown through data access and parsing; mapping to a knowledge body structure through entity extraction, relationship identification, and semantic annotation.

2)Adaptive Semantic Chunking Algorithm

To address the semantic fragmentation caused by traditional fixed-length chunking, an adaptive chunking algorithm based on semantic coherence analysis is proposed. The algorithm determines chunk boundaries by calculating the semantic difference between adjacent sentences:

$$y = 1 - \cos ine - similarity(E_{i-1}, E_i) \tag{1}$$

where $\boldsymbol{E_i}$ represents the embedding vector of the i-th sentence. The chunking threshold is adaptively calculated as:

$$\alpha = \max(0.1, N/(N-K)) \tag{2}$$

where N is the total number of sentences, and K is the base number of chunks.

3)Multi-Index Construction

Two types of indexes are constructed: keyword inverted index and semantic vector index. The semantic vector index encodes knowledge fragments using the Sentence-BERT model and builds an approximate nearest neighbor search via the HNSW algorithm. The keyword index uses ElasticSearch for fast matching.

C. Prompt Engineering-Driven Semantic Alignment Optimization Mechanism

1)Semantic Alignment Optimization Framework

To address the semantic space mismatch between queries and the knowledge base, a prompt engineering-based alignment optimization method is proposed. Using the 5W1H (What, Why, Where, When, Who, How) framework, hypothetical question-answer pairs are generated to extend knowledge fragments. The specific process is as follows: use prompt templates to generate multi-dimensional question-answer pairs; concatenate the original passage with the generated question-

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answer pairs; unify the semantic space through vectorized representation. This method achieves alignment without fine-tuning.

2)Multi-Dimensional Prompt Template Design

The prompt template is designed based on the 5W1H framework, comprehensively covering user query intent from six dimensions. Each dimension generates 3-5 question-answer pairs, significantly improving retrieval recall. Table 1 shows the specific design of the prompt template.

TABLE I. SEMANTIC ALIGNMENT OPTIMIZATION PROMPT TEMPLATE

Dimension	Prompt Template	Generation Example
What	Based on the passage [content], generate a question about "what"	What is dynamic knowledge retrieval?
Why	Generate a "why" type question based on causal explanations in the passage.	Why is semantic alignment needed?
Where	Generate questions related to the passage from a geographical dimension	In which scenarios is dynamic indexing applied?
When	Generate questions based on temporal information	How frequently is the knowledge base updated?
Who	Generate questions involving subjects or entities	Who is responsible for knowledge base maintenance?
How	Generate questions about processes or methods	How to implement semantic chunking?

D. Intelligent Routing Decision Model

The system employs a lightweight neural network to dynamically select between retrieval and generation paths. The decision model extracts features from user input, including word vectors, interrogative structures, and historical behavior features, and calculates path probabilities using a dual-tower structure. If the retrieval probability exceeds a threshold , the knowledge retrieval process is triggered. The routing decision formula is:

$$P_{retrieve} = \sigma\left(W\left[P_{query}, P_{history}\right] + b\right) \tag{3}$$

where σ is the sigmoid function, f_{query} is the query feature, and f_{history} is the historical feature. The model supports multi-feature fusion and dialogue state tracking.

E. Hybrid Retrieval and Re-Ranking Mechanism

1)Multi-Feature Fusion Retrieval

The retrieval phase combines sparse retrieval (BM25) and dense retrieval (DPR) through a weighted score for hybrid retrieval. The BM25 score is calculated as:

$$BM25(q,d) = \sum IDF(q_i) \cdot \frac{f(q_i,d)\cdot(k_1+1)}{f(q_i,D) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{avgdl}\right)} \tag{4}$$

where q represents a query, d represents a document, q_i represents the i-th word in the query, IDF(q_i) denotes the inverse document frequency of word q_i, which measures the importance of the word. $f(q_i, \text{d})$ is the term frequency of q_i in document d, k_1 and b are hyperparameters; $|d|$ is the length of document d and $avgdl$ represents the average length of all documents.

The hybrid score is calculated as:

$$score_{hybrid} = \lambda score_{BM25} + (1-\lambda) score_{dense} \tag{5}$$

where λ is a balancing parameter dynamically adjusted based on query complexity. The balancing parameter λ is dynamically adjusted based on query complexity. For short and simple queries (length < 10 words and containing fewer than 2 entities), λ is set to 0.7 to prioritize the lexical matching strength of BM25. For long or complex queries (length $\geqslant$ 15 words or containing 4 or more entities), λ is set to 0.3 to emphasize the semantic matching capability of dense retrieval. For all other queries, a balanced value of $\lambda = 0.5$ is used. The parameter selection is based on empirical optimization from our preliminary experiments involving 500 test cases with different query characteristics.

2) Hierarchical Re-Ranking

The re-ranking phase uses a cross-encoder to semantically score candidate results, combining semantic similarity, context adaptation, and knowledge source credibility for comprehensive evaluation. The final ranking is optimized using the LambdaMART model to improve the accuracy of top-k results.

IV. Experimental Results Analysis

A. Experimental Setup

To evaluate the effectiveness of the proposed method, experiments were conducted on two widely used open-domain QA benchmark datasets: HotpotQA [13] and SQuAD [14]. HotpotQA focuses on multi-hop reasoning tasks, containing approximately 113,000 QA instances requiring models to integrate information from multiple documents for complex reasoning. In the distractor setting, each question is accompanied by 10 documents, of which only 2 are relevant resources and the remaining 8 are distractors. The FullWiki setting provides only the Wikipedia full text and the question itself. Since the test set is not publicly available, the development set under the FullWiki setting was used for performance evaluation. SQuAD is a single-hop QA dataset built from 536 English Wikipedia articles, containing over 107,000 question-answer pairs where the answer to each question is directly extracted from the corresponding article.

The experimental setup involved 500 test cases. Comparison methods included traditional RAG methods and Self-RAG[15]. Evaluation metrics included exact match (EM) and fuzzy matching(F1).The hardware environment consisted of 4 Tesla A800 GPUs, using Llama2-7b as the generative model.

B. Main Results Analysis

Experimental results listed in TABLE 2 demonstrate that the proposed method achieves highly competitive performance on both datasets. Compared to Self-RAG, our method achieves 9.65% and 9.62% improvement in EM and F1 scores on

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SQuAD, respectively. This significant enhancement demonstrates the effectiveness of our dynamic indexing and semantic alignment components in addressing semantic fragmentation issues prevalent in traditional methods.On the SQuAD dataset, our method outperforms all other approaches in both Exact Match (EM) and F1 scores, showing significant improvements over the second-best method, Self-RAG. This indicates that our approach more effectively retrieves and utilizes relevant contextual information when handling extractive question answering tasks. On the HotpotQA dataset, our method achieves EM and F1 scores of 37.01 and 47.53, respectively, surpassing all other comparative methods and demonstrating particular advantages in processing multi-hop reasoning questions.

TABLE II. PERFORMANCE COMPARISON OF DIFFERENT METHODS ON HOTPOTQA AND SQUAD DATASETS

Method	HotpotQA		SQuAD	
	EM	F1	EM	F1
Traditional RAG	23.06	32.22	13.40	23.10
Self-RAG	34.38	45.97	25.47	36.71
OURS	37.01	47.53	35.12	45.69

C. *Ablation Study Analysis*

Ablation experiments were conducted to verify the contribution of each module . As shown in TABLE III, results show that the dynamic semantic chunking module significantly improves long document processing; its removal causes the EM score to drop by 13.04%. The semantic alignment optimization mechanism improves retrieval precision by 22.2%. These results confirm the necessity of synergistic work among the modules.

TABLE III. ABLATION EXPERIMENT RESULTS

Module Combination	EM	F1	Retrieval Precision
Full System	37.01	47.53	90.1%
W/O Dynamic Chunking	32.74	42.03	78.7%
W/O Semantic Alignment	30.09	39.68	76.3%

V. CONCLUSIONS

The retrieval-augmented generation method based on dynamic knowledge indexing and semantic alignment optimization proposed in this paper, through the effective collaboration of the dynamic multi-granularity indexing architecture, the prompt engineering-driven semantic alignment mechanism, and the intelligent routing decision model, has significantly improved retrieval accuracy and hallucination suppression rates in knowledge-intensive tasks. This empirically demonstrates its practicality and scalability in open-domain question answering and vertical domain applications.

However, this study still has limitations, including inherent technical constraints: The semantic alignment mechanism relies heavily on the quality of generated question-answer pairs, which may introduce bias when handling domain-specific terminology. Additionally, the dynamic indexing architecture requires substantial computational resources during real-time updates, potentially limiting deployment in resource-constrained environments.Future work will focus on: optimizing the dynamic update mechanism of the knowledge base by introducing incremental learning and stream processing to enhance timeliness; expanding multimodal fusion capabilities by integrating visual-language models to support non-textual data parsing; deepening domain adaptation techniques by combining meta-learning to achieve rapid migration to low-resource scenarios; refining the collaborative optimization of the intelligent routing and generation models by using reinforcement learning to improve system adaptability; and establishing a multi-dimensional evaluation system and an ethical security framework to promote the reliable deployment of the technology in practical scenarios.

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